Extended RPA in the particle–hole, particle–particle and crossed particle–hole channels for an arbitrary vacuum: reduced-density-matrix form and active/inactive block structure

The three single-channel random-phase problems of the fixed-kernel channel model of `Pa2CCD.tex` (direct electron–hole, transverse or crossed-ring electron–hole, and particle–particle) are written in their extended form, *i.e.* as Rowe equations of motion on an arbitrary correlated vacuum $|\Psi_0\rangle$. All matrix elements are exact functionals of the one- and two-particle reduced density matrices of the vacuum only. Compact closed forms are given in terms of the generalized Fock matrix, the energy-weighted density (Lagrangian) matrix, the antisymmetrized integrals and the 2-RDM or its cumulant. For a CAS-type vacuum (closed core ⊗ correlated active space ⊗ empty virtuals) every matrix is then resolved into the blocks that carry inactive indices only, active indices only, and mixed indices. All formulas have been verified numerically against brute-force Fock-space double commutators.

I. SCOPE AND CONVENTIONS

`Pa2CCD.tex` reduces the parquet channel equations with fixed bare kernels to three static random-phase problems, one per correlation channel (direct electron–hole eh, transverse electron–hole $\overline{\text{eh}}$, particle–particle pp), all built on a Hartree–Fock determinant. These notes give the corresponding equations for an arbitrary vacuum $|\Psi_0\rangle$ in the "extended" form of Rowe's equation-of-motion method[1,2]: the pair operators of each channel act on the correlated vacuum, the double commutator with $\hat{H}$ replaces the bare kernel, and the commutator metric replaces the Hartree–Fock pair norm. Every matrix element is then an exact functional of the one- and two-particle reduced density matrices (RDMs) of $|\Psi_0\rangle$. The second part (Sec. VI) resolves these matrices into inactive, active and mixed blocks for the CAS-type vacuum used throughout `mr-hedin.tex`.

Hamiltonian and integrals. All quantities are real (real spin orbitals, real vacuum). With $\langle pq|rs\rangle = (pr|qs) = v_{pr,qs}$ the notation of `mr-hedin.tex`,

$$\hat{H} = \sum_{pq} h_{pq}\,\hat{a}_p^\dagger \hat{a}_q + \frac{1}{2}\sum_{pqrs}\langle pq|rs\rangle\,\hat{a}_p^\dagger\hat{a}_q^\dagger\hat{a}_s\hat{a}_r,$$
$$\langle pq||rs\rangle = \langle pq|rs\rangle - \langle pq|sr\rangle = v_{pr,qs} - v_{ps,qr}. \tag{1}$$

General spin orbitals are p, q, r, s and general summation indices t, u; in Sec. VI i, j, k, l denote core, a, b, c, d virtual and x, y, z, w active spin orbitals, and the summation indices u, v run over active spin orbitals only.

Reduced density matrices.

$$\gamma_{pq} = \langle\Psi_0|\hat{a}_p^\dagger\hat{a}_q|\Psi_0\rangle, \qquad \Gamma_{pq,rs} = \langle\Psi_0|\hat{a}_p^\dagger\hat{a}_q^\dagger\hat{a}_s\hat{a}_r|\Psi_0\rangle, \tag{2}$$
$$\lambda_{pq,rs} = \Gamma_{pq,rs} - \gamma_{pr}\gamma_{qs} + \gamma_{ps}\gamma_{qr}, \tag{3}$$

so that $E_0 = \sum_{pq} h_{pq}\gamma_{pq} + \frac{1}{2}\sum v_{pr,qs}\Gamma_{pq,rs}$, $\Gamma_{pq,rs} = -\Gamma_{qp,rs} = -\Gamma_{pq,sr} = \Gamma_{rs,pq}$ and λ has the same symmetries. Γ coincides with $D^{(2)}$ of `numerics/equations.tex` [its Eq. (rdm-def)] and with $\Gamma_{pq,sr}$ in the notation of the EKT section of `mr-hedin.tex`. Nothing is assumed about $|\Psi_0\rangle$ except that it is a normalized N-electron state; in particular it need not be an eigenstate of $\hat{H}$.

Two one-body matrices. All closed forms below are expressed with the generalized Fock matrix f and the energy-weighted density (Lagrangian) matrix $\mathcal{L}$,

$$f_{pq} = h_{pq} + \sum_{tu}\gamma_{tu}\,\langle pt||qu\rangle, \tag{4}$$
$$\mathcal{L}_{pq} = \sum_t h_{pt}\gamma_{tq} + \frac{1}{2}\sum_{tuv}\langle pu||tv\rangle\,\Gamma_{qu,tv}$$
$$= \sum_t f_{pt}\gamma_{tq} + \frac{1}{2}\sum_{tuv}\langle pu||tv\rangle\,\lambda_{qu,tv}, \tag{5}$$
$$\bar{\mathcal{L}}_{pq} = \tfrac{1}{2}(\mathcal{L}_{pq} + \mathcal{L}_{qp}). \tag{6}$$

f is symmetric; $\mathcal{L}$ is symmetric only if $|\Psi_0\rangle$ is stationary with respect to orbital rotations, because

$$\langle\Psi_0|[\hat{H}, \hat{a}_p^\dagger\hat{a}_q]|\Psi_0\rangle = \mathcal{L}_{pq} - \mathcal{L}_{qp} \tag{7}$$

(generalized Brillouin condition). $\mathrm{Tr}\,\mathcal{L} = E_1 + 2E_2$.

Status markers. As in `mr-hedin.tex`: **Exact identity.** exact identities; **Established result.** results available in the literature or in the other notes; **Proposed approximation.** definitions made here.

II. EQUATION OF MOTION IN AN OPERATOR MANIFOLD

Established result. For a set of (even) operators $\{\hat{Q}_\mu^\dagger\}$ define

$$\mathcal{M}_{\mu\nu} = \langle\Psi_0|[\hat{Q}_\mu, [\hat{H}, \hat{Q}_\nu^\dagger]]|\Psi_0\rangle, \qquad \mathcal{S}_{\mu\nu} = \langle\Psi_0|[\hat{Q}_\mu, \hat{Q}_\nu^\dagger]|\Psi_0\rangle, \tag{8}$$
$$\boldsymbol{\mathcal{M}}\,c_\nu = \omega_\nu\,\boldsymbol{\mathcal{S}}\,c_\nu, \qquad c_\nu^T\boldsymbol{\mathcal{S}}c_{\nu'} = \pm\delta_{\nu\nu'}. \tag{9}$$

If $|\Psi_0\rangle$ is the exact ground state and the manifold contains the exact excitation operators $|n\rangle\langle 0|$, the solutions of Eq. (9) are the exact excitation energies and the positive-norm (negative-norm) eigenvectors give the exact transition amplitudes $\langle n|\hat{Q}_\nu^\dagger|0\rangle$ (de-excitation amplitudes); for a truncated manifold Eq. (9) is the inner projection of the superoperator resolvent onto the manifold, and the corresponding pair propagator is

$$\boldsymbol{\Pi}(\Omega) = \boldsymbol{\mathcal{S}}\,(\Omega\boldsymbol{\mathcal{S}} - \boldsymbol{\mathcal{M}})^{-1}\boldsymbol{\mathcal{S}} = \sum_\nu \frac{(\boldsymbol{\mathcal{S}}c_\nu)(\boldsymbol{\mathcal{S}}c_\nu)^T}{c_\nu^T\boldsymbol{\mathcal{S}}c_\nu\,(\Omega - \omega_\nu)}. \tag{10}$$

Equation (10) is the extended-vacuum counterpart of the static channel propagators $\boldsymbol{L}^\alpha(\Omega) = \mathrm{i}\sigma_\alpha(\Omega\boldsymbol{\eta}^\alpha - \boldsymbol{\mathcal{H}}^\alpha)^{-1}$ of

`Pa2CCD.tex`: $\boldsymbol{\mathcal{S}}$ plays the role of the sector metric $\boldsymbol{\eta}^{\alpha}$ (it is no longer $\pm\mathbf{1}$) and $\boldsymbol{\mathcal{M}}$ that of $\boldsymbol{\mathcal{H}}^{\alpha} = \boldsymbol{\mathcal{H}}_0^{\alpha} + \boldsymbol{V}^{\alpha}$.

Exact identity. For a vacuum that is not an eigenstate of $\hat{H}$, the double commutator (8) is not symmetric; by the Jacobi identity $\mathcal{M}_{\mu\nu} - \mathcal{M}_{\nu\mu} = \langle[\hat{H},[\hat{Q}_\mu,\hat{Q}_\nu^\dagger]]\rangle$, which for the pair manifolds below reduces to Kronecker deltas times $\mathcal{L} - \mathcal{L}^T$ [see Eq. (15)]. Throughout, $\boldsymbol{\mathcal{M}}$ denotes the symmetric part $\frac{1}{2}(\boldsymbol{\mathcal{M}} + \boldsymbol{\mathcal{M}}^T)$, which is what the implementation in `numerics/mrgw_hermitian/ekt_erpa.py` uses; in all closed forms this amounts to replacing $\mathcal{L}$ by $\bar{\mathcal{L}}$.

III. PARTICLE–HOLE CHANNEL (eh, RING)

Established result. The manifold consists of all single replacements $\hat{E}_{rs} = \hat{a}_r^\dagger \hat{a}_s$, $r \neq s$, on the correlated vacuum; the composite index (pq) labels the bra operator $\hat{E}_{pq}$, whose adjoint $\hat{E}_{qp}$ appears in the commutators, *i.e.*

$$\mathcal{M}_{(pq),(rs)} = \langle[\hat{E}_{qp},[\hat{H},\hat{E}_{rs}]]\rangle, \qquad \mathcal{S}_{(pq),(rs)} = \langle[\hat{E}_{qp},\hat{E}_{rs}]\rangle. \tag{11}$$

This is the ERPA of Chatterjee and Pernal[2] and the active-space ERPA of `mr-hedin.tex` [its Eq. (erpa-eom)], here written in the full orbital space.

Exact identity. The metric and the (symmetrized) double commutator are

$$\mathcal{S}_{(pq),(rs)} = \delta_{pr}\gamma_{qs} - \delta_{qs}\gamma_{pr}, \tag{12}$$

$$\mathcal{M}_{(pq),(rs)} = h_{pr}\gamma_{qs} + h_{qs}\gamma_{pr} - \delta_{pr}\bar{\mathcal{L}}_{qs} - \delta_{qs}\bar{\mathcal{L}}_{pr} + \sum_{tu}\Big[\langle pt\|ur\rangle\,\Gamma_{qt,us} + \langle qt\|us\rangle\,\Gamma_{pt,ur}\Big] - \frac{1}{2}\sum_{tu}\Big[\langle ps\|tu\rangle\,\Gamma_{qr,tu} + \langle qr\|tu\rangle\,\Gamma_{ps,tu}\Big] \tag{13}$$

$$= \gamma_{qs}f_{pr} + \gamma_{pr}f_{qs} - \delta_{pr}\bar{\mathcal{L}}_{qs} - \delta_{qs}\bar{\mathcal{L}}_{pr} + \sum_{tu}\Big[\langle pt\|ur\rangle\gamma_{qu}\gamma_{ts} + \langle qt\|us\rangle\gamma_{pu}\gamma_{tr} - \langle ps\|tu\rangle\gamma_{qt}\gamma_{ru} - \langle qr\|tu\rangle\gamma_{pt}\gamma_{su}\Big]$$
$$+ \sum_{tu}\Big[\langle pt\|ur\rangle\lambda_{qt,us} + \langle qt\|us\rangle\lambda_{pt,ur}\Big] - \frac{1}{2}\sum_{tu}\Big[\langle ps\|tu\rangle\lambda_{qr,tu} + \langle qr\|tu\rangle\lambda_{ps,tu}\Big]. \tag{14}$$

Equation (13) is the 20-term list of `numerics/rdm_terms.tex` (generated by `wick.py`) after collecting the direct and exchange integrals; the unsymmetrized double commutator is obtained by replacing $\bar{\mathcal{L}}_{qs} \to \mathcal{L}_{sq}$ and $\bar{\mathcal{L}}_{pr} \to \mathcal{L}_{rp}$, so that

$$\mathcal{M}_{(pq),(rs)} - \mathcal{M}_{(rs),(pq)} = -\delta_{pr}(\mathcal{L}_{sq} - \mathcal{L}_{qs}) - \delta_{qs}(\mathcal{L}_{rp} - \mathcal{L}_{pr}) \tag{15}$$

before symmetrization. The cumulant form (14) makes the mean-field content explicit: for a single determinant ($\lambda = 0$, $\gamma^2 = \gamma$) it collapses to the usual RPA matrices, see below. Two further exact properties, both used in the next paragraph, are $\mathcal{M}_{(qp),(sr)} = \mathcal{M}_{(pq),(rs)}$ (adjoint of the double commutator with real quantities) and the block-diagonality of $\boldsymbol{\mathcal{S}}$ in the natural-orbital basis.

Established result. *RPA form.* In the natural spin-orbital basis $\gamma = \mathrm{diag}(n_p)$ the metric is diagonal, $\mathcal{S}_{(pq),(rs)} = \delta_{pr}\delta_{qs}(n_q - n_p)$. Let $\mathcal{E} = \{(pq) : n_q > n_p\}$ be the excitation pairs (particle-like index first), $\bar{\mathcal{E}} = \{(qp)\}$ their de-excitation partners, and $N_{(pq)} = n_q - n_p > 0$. With

$$A_{(pq),(rs)} = \mathcal{M}_{(pq),(rs)}, \quad B_{(pq),(rs)} = -\mathcal{M}_{(pq),(sr)}, \quad (pq),(rs) \in \mathcal{E}, \tag{16}$$

both symmetric, Eq. (9) with $\hat{Q}_\nu^\dagger = \sum_{\mathcal{E}}(X_{\nu,pq}\hat{E}_{pq} - Y_{\nu,pq}\hat{E}_{qp})$ becomes

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{A} \end{pmatrix}\begin{pmatrix} \boldsymbol{X}_\nu \\ \boldsymbol{Y}_\nu \end{pmatrix} = \omega_\nu \begin{pmatrix} \boldsymbol{N} & 0 \\ 0 & -\boldsymbol{N} \end{pmatrix}\begin{pmatrix} \boldsymbol{X}_\nu \\ \boldsymbol{Y}_\nu \end{pmatrix}, \tag{17}$$

i.e. the standard RPA structure with the metric $\boldsymbol{N}$; the transition densities are $\rho^\nu_{pq} = N_{(pq)}X_{\nu,pq}$ and $\rho^\nu_{qp} = -N_{(pq)}Y_{\nu,pq}$ [Eq. (erpa-rho) of `mr-hedin.tex`, whose $\bar{B}$ equals $-\boldsymbol{B}$ because its de-excitation coefficients are $-\boldsymbol{Y}$]. Pairs with $n_p = n_q$ have zero norm and are removed from the manifold. For a Hartree–Fock vacuum $\mathcal{E} = \{(ai)\}$ and Eqs. (14) and (16) give $A_{(ai),(bj)} = \delta_{ij}f_{ab} - \delta_{ab}f_{ij} + \langle aj\|ib\rangle$, $B_{(ai),(bj)} = \langle ab\|ij\rangle$, the matrices of the direct electron–hole channel of `Pa2CCD.tex` [its $\boldsymbol{A}^{\mathrm{eh}}$, $\boldsymbol{B}^{\mathrm{eh}}$].

IV. PARTICLE–PARTICLE CHANNEL (pp, LADDER)

Established result. The manifold consists of all pair-creation operators $\hat{a}_r^\dagger\hat{a}_s^\dagger$, $r < s$, on the correlated vacuum (no distinction between particle and hole pairs is made at this stage):

$$\mathcal{M}^{\mathrm{pp}}_{(pq),(rs)} = \langle[\hat{a}_q\hat{a}_p,[\hat{H},\hat{a}_r^\dagger\hat{a}_s^\dagger]]\rangle, \qquad \mathcal{S}^{\mathrm{pp}}_{(pq),(rs)} = \langle[\hat{a}_q\hat{a}_p,\hat{a}_r^\dagger\hat{a}_s^\dagger]\rangle. \tag{18}$$

Both are antisymmetric under $p \leftrightarrow q$ and under $r \leftrightarrow s$. With $P(pq)X_{pq} = X_{pq} - X_{qp}$,

Exact identity.

$$\mathcal{S}^{\rm pp}_{(pq),(rs)} = \delta_{pr}\delta_{qs} - \delta_{ps}\delta_{qr} - \delta_{pr}\gamma_{qs} + \delta_{ps}\gamma_{qr} + \delta_{qr}\gamma_{ps} - \delta_{qs}\gamma_{pr}, \tag{19}$$

$$\begin{aligned}\mathcal{M}^{\rm pp}_{(pq),(rs)} &= P(pq)P(rs)\Big[\delta_{qs}(f_{pr} - \bar{\mathcal{L}}_{pr}) - h_{pr}\gamma_{qs} + \sum_{tu}\langle pt\|ru\rangle\,\Gamma_{ts,qu}\Big] \\ &\quad + \langle pq\|rs\rangle - \sum_{t}\Big[\gamma_{pt}\langle tq\|rs\rangle + \gamma_{qt}\langle pt\|rs\rangle + \gamma_{rt}\langle pq\|ts\rangle + \gamma_{st}\langle pq\|rt\rangle\Big] \qquad (20)\\ &= P(pq)P(rs)\Big[\delta_{qs}(f_{pr} - \bar{\mathcal{L}}_{pr}) - \gamma_{qs}f_{pr} + \sum_{tu}\langle pt\|ru\rangle\gamma_{qt}\gamma_{su} + \sum_{tu}\langle pt\|ru\rangle\lambda_{ts,qu}\Big] \\ &\quad + \langle pq\|rs\rangle - \sum_{t}\Big[\gamma_{pt}\langle tq\|rs\rangle + \gamma_{qt}\langle pt\|rs\rangle + \gamma_{rt}\langle pq\|ts\rangle + \gamma_{st}\langle pq\|rt\rangle\Big]. \qquad (21)\end{aligned}$$

Equation (20) collects the 72 terms produced by `wick.py` for the pair manifold; the unsymmetrized double commutator has $\mathcal{L}_{rp}$ in place of $\bar{\mathcal{L}}_{pr}$. Note that the 2-RDM enters only through particle–hole-type contractions $\sum_{tu}\langle pt\|ru\rangle\Gamma_{ts,qu}$ and through $\mathcal{L}$; the ladder-type integral $\langle pq\|rs\rangle$ is dressed by the 1-RDM only (one factor $-\gamma$ on each of the four legs).

Established result. *RPA form.* In the natural spin-orbital basis, for $p<q$ and $r<s$, $\mathcal{S}^{\rm pp}_{(pq),(rs)} = \delta_{pr}\delta_{qs}(1-n_p-n_q)$. Pairs with $1-n_p-n_q>0$ form the particle-like sector $\mathcal{P}$, pairs with $1-n_p-n_q<0$ the hole-like sector $\mathcal{H}$, and pairs with $n_p+n_q=1$ (*e.g.* one fully occupied and one empty orbital) have zero norm and are removed. With $N^{\rm pp}_{(pq)} = |1-n_p-n_q|$ and

$$\boldsymbol{A}^{\rm pp} = \boldsymbol{\mathcal{M}}^{\rm pp}_{\mathcal{PP}}, \qquad \boldsymbol{C}^{\rm pp} = \boldsymbol{\mathcal{M}}^{\rm pp}_{\mathcal{HH}}, \qquad \boldsymbol{B}^{\rm pp} = -\boldsymbol{\mathcal{M}}^{\rm pp}_{\mathcal{PH}}, \tag{22}$$

Eq. (9) with $\hat{Q}^\dagger_\nu = \sum_{\mathcal{P}} X_{\nu,pq}\hat{a}^\dagger_p\hat{a}^\dagger_q - \sum_{\mathcal{H}} Y_{\nu,pq}\hat{a}^\dagger_p\hat{a}^\dagger_q$ reads

$$\begin{pmatrix}\boldsymbol{A}^{\rm pp} & \boldsymbol{B}^{\rm pp}\\ \boldsymbol{B}^{\rm pp\,T} & \boldsymbol{C}^{\rm pp}\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_\nu\\ \boldsymbol{Y}_\nu\end{pmatrix} = \omega_\nu\begin{pmatrix}\boldsymbol{N}^{\rm pp}_{\mathcal{P}} & 0\\ 0 & -\boldsymbol{N}^{\rm pp}_{\mathcal{H}}\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_\nu\\ \boldsymbol{Y}_\nu\end{pmatrix}. \tag{23}$$

Positive-norm solutions are the $N{+}2$ states, $\omega_\nu = E^{N+2}_\nu - E_0$; negative-norm solutions are the $N{-}2$ states with $\omega_\nu = -(E^{N-2}_\nu - E_0)$[3,4]. The sign in the definition of $\boldsymbol{B}^{\rm pp}$ compensates the sign of the commutator metric in $\mathcal{H}$ and makes the Hartree–Fock limit coincide with the particle–particle channel of `Pa2CCD.tex`: $\mathcal{P} = \{(ab)\}$, $\mathcal{H} = \{(ij)\}$ and Eq. (21) gives, in canonical orbitals, $A^{\rm pp}_{ab,cd} = (\epsilon_a+\epsilon_b)\delta^{\rm A}_{ab,cd} + \langle ab\|cd\rangle$, $C^{\rm pp}_{ij,kl} = -(\epsilon_i+\epsilon_j)\delta^{\rm A}_{ij,kl} + \langle ij\|kl\rangle$, $B^{\rm pp}_{ab,ij} = \langle ab\|ij\rangle$ [its $\boldsymbol{C}^{\rm pp}$, $\boldsymbol{D}^{\rm pp}$, $\boldsymbol{B}^{\rm pp}$ with ordered pairs].

V. CROSSED-RING PARTICLE–HOLE CHANNEL ($\overline{\rm eh}$)

Established result. The transverse channel is the electron–hole channel with crossed external labels: the composite pair joins the hole label of one external vertex with the particle label of the other, and the fermionic crossing sign $\hat{E}_{bi}\hat{E}_{aj}|\Phi_0\rangle = -\hat{E}_{ai}\hat{E}_{bj}|\Phi_0\rangle$ [Eq. (crossed_operator_sign) of `Pa2CCD.tex`] is carried by the forward–backward blocks. In the Hartree–Fock static channel reduction of `Pa2CCD.tex` the generator in the crossed composite ordering $(ib),(ja)$ is

$$\boldsymbol{H}^{\overline{\rm eh}} = \begin{pmatrix}\boldsymbol{A}^{\overline{\rm eh}} & -\boldsymbol{B}^{\overline{\rm eh}}\\ \boldsymbol{B}^{\overline{\rm eh}\dagger} & -\boldsymbol{A}^{\overline{\rm eh}\dagger}\end{pmatrix},$$

$$\boldsymbol{A}^{\overline{\rm eh}}_{ib,kc} = (\epsilon_b-\epsilon_i)\delta_{ik}\delta_{bc} + \langle bk\|ic\rangle, \qquad \boldsymbol{B}^{\overline{\rm eh}}_{ib,ja} = \langle ba\|ij\rangle. \tag{24}$$

Read as arrays over (hole, particle) pairs, $\boldsymbol{A}^{\overline{\rm eh}}$ and $\boldsymbol{B}^{\overline{\rm eh}}$ are identical to the direct-channel matrices $\boldsymbol{A}^{\rm eh}_{ia,jb} = (\epsilon_a-\epsilon_i)\delta_{ij}\delta_{ab} + \langle aj\|ib\rangle$ and $\boldsymbol{B}^{\rm eh}_{ia,jb} = \langle ab\|ij\rangle$; the crossing is carried entirely by the sign of the forward–backward block and by the crossed attachment of the composite indices to the external legs (the map $J_{\overline{\rm eh}}$ of `Pa2CCD.tex`). With $\boldsymbol{Z} = {\rm diag}(\boldsymbol{1},-\boldsymbol{1})$ on the forward/backward sectors, $\boldsymbol{H}^{\overline{\rm eh}} = \boldsymbol{Z}\boldsymbol{H}^{\rm eh}\boldsymbol{Z}$ and $\boldsymbol{L}^{\overline{\rm eh}}(\Omega) = \boldsymbol{Z}\boldsymbol{L}^{\rm eh}(\Omega)\boldsymbol{Z}$: the crossed channel has the same poles as the direct one, the eigenvectors $(\boldsymbol{X}_\nu,-\boldsymbol{Y}_\nu)$, and its static propagator differs from the direct one by the sign of the forward–backward blocks. The channels differ once they are attached to external legs; in the common-amplitude construction this is what turns the ring terms of the eh channel into the crossed-ring terms of the $\overline{\rm eh}$ channel [Eqs. (ring_linear_audit) and (ring_quadratic_audit) of `Pa2CCD.tex`].

Proposed approximation. *Extended form.* The same construction defines the crossed-ring channel on an arbitrary vacuum. Let $\boldsymbol{Z} = \boldsymbol{\mathcal{S}}\,(\boldsymbol{\mathcal{S}}^2)^{-1/2}$ be the sign of the particle–hole metric on its nonzero-norm subspace ($\boldsymbol{Z} = {\rm diag}(\boldsymbol{1},-\boldsymbol{1})$ on $\mathcal{E}\oplus\bar{\mathcal{E}}$ in the natural-orbital basis). The crossed-ring ERPA is

$$\boldsymbol{\mathcal{M}}^{\overline{\rm eh}} = \boldsymbol{Z}\,\boldsymbol{\mathcal{M}}\,\boldsymbol{Z}, \qquad \boldsymbol{\mathcal{S}}^{\overline{\rm eh}} = \boldsymbol{Z}\boldsymbol{\mathcal{S}}\boldsymbol{Z} = \boldsymbol{\mathcal{S}},$$
$$\begin{pmatrix}\boldsymbol{A} & -\boldsymbol{B}\\ -\boldsymbol{B} & \boldsymbol{A}\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_\nu\\ \boldsymbol{Y}_\nu\end{pmatrix} = \omega_\nu\begin{pmatrix}\boldsymbol{N} & 0\\ 0 & -\boldsymbol{N}\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_\nu\\ \boldsymbol{Y}_\nu\end{pmatrix}, \tag{25}$$

with $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{N}$ of Eqs. (16)–(17), and its propagator is $\boldsymbol{\Pi}^{\overline{\rm eh}}(\Omega) = \boldsymbol{Z}\boldsymbol{\Pi}(\Omega)\boldsymbol{Z}$. No matrix elements beyond those of Sec. III are needed: the 1-/2-RDM content of the crossed channel is that of Eqs. (12)–(14), with the sign of $\boldsymbol{B}$ reversed. Equation (25) reduces to Eq. (24) for a Hartree–Fock vacuum and is the unique extension for which the exact crossing relation $L(12;1'2') = -L(12;2'1')$ of `parquet.tex` acts on the static channel propagator in the same way as in the Hartree–Fock case, namely on the forward/backward structure only. (With the antisymmetrized kernel both the direct and the exchange

integrals are present in every channel; "ring" and "crossed ring" refer to the pairing of the external legs, not to a direct/exchange split of the kernel.)

VI. BLOCK STRUCTURE FOR A CAS-TYPE VACUUM

A. Structure of the vacuum and of f, $\mathcal{L}$

Exact identity. Let $|\Psi_0\rangle = |\Phi_{\rm core}\rangle \otimes |\Xi\rangle$ with all core spin orbitals $i, j, \ldots$ occupied, all virtual spin orbitals $a, b, \ldots$ empty, and $|\Xi\rangle$ an *arbitrary* $N_{\rm act}$-electron state in the active spin orbitals $x, y, \ldots$ (CASCI or CASSCF, but no stationarity is assumed). Then

$$\begin{aligned} &\gamma_{ij} = \delta_{ij}, \quad \gamma_{ab} = 0, \quad \gamma_{xy} = \gamma^{\rm act}_{xy}, \quad \gamma_{Px} = \gamma_{ia} = 0, \\ &\lambda_{pq,rs} \neq 0 \quad \text{only for } p,q,r,s \in \text{active}, \end{aligned} \tag{26}$$

and Γ follows from Eq. (3). Inserting Eq. (26) into Eqs. (4)–(6), with u, v, z active,

$$f_{pq} = h_{pq} + \sum_k \langle pk||qk\rangle + \sum_{uv} \gamma_{uv}\langle pu||qv\rangle, \tag{27}$$

$$\mathcal{L}_{pj} = f_{pj}, \qquad \mathcal{L}_{pb} = 0, \qquad \mathcal{L}_{py} = \sum_u f_{pu}\gamma_{uy} + \Lambda_{py},$$

$$\Lambda_{py} = \frac{1}{2}\sum_{uvz} \langle pu||vz\rangle\, \lambda_{yu,vz}, \tag{28}$$

so that

$$\begin{aligned} &\bar{\mathcal{L}}_{ij} = f_{ij}, \qquad \bar{\mathcal{L}}_{ab} = 0, \qquad \bar{\mathcal{L}}_{ia} = \tfrac{1}{2} f_{ia}, \\ &\bar{\mathcal{L}}_{ix} = \tfrac{1}{2}\Big[f_{xi} + \sum_u f_{iu}\gamma_{ux} + \Lambda_{ix}\Big], \\ &\bar{\mathcal{L}}_{ax} = \tfrac{1}{2}\Big[\sum_u f_{au}\gamma_{ux} + \Lambda_{ax}\Big], \\ &\bar{\mathcal{L}}_{xy} = \tfrac{1}{2}\Big[\sum_u (f_{xu}\gamma_{uy} + f_{yu}\gamma_{ux}) + \Lambda_{xy} + \Lambda_{yx}\Big]. \end{aligned} \tag{29}$$

f is the generalized (Dyall) Fock matrix that also defines the inactive orbital energies $\kappa_P = \epsilon_P$ in `mr-hedin.tex`, and its active block is $f_{xy} = h^{\rm eff}_{xy} + \sum_{uv}\gamma_{uv}\langle xu||yv\rangle$ with $h^{\rm eff}_{xy} = h_{xy} + \sum_k \langle xk||yk\rangle$ the one-body part of the active Dyall Hamiltonian; $\mathcal{L}_{xy}$ is therefore the Lagrangian of the active-space problem. For a CASSCF vacuum $\mathcal{L}$ is symmetric [Eq. (7)]: $f_{ia} = 0$, $f_{xi} = \sum_u f_{iu}\gamma_{ux} + \Lambda_{ix}$, $\sum_u f_{au}\gamma_{ux} + \Lambda_{ax} = 0$, and the bars in Eq. (29) can be dropped; for CASCI orbitals the symmetrization is part of the definition of $\boldsymbol{\mathcal{M}}$.

In the block formulas below every summation index u, v runs over the *active* spin orbitals only: the core contractions have been absorbed into f and the Kronecker deltas, and the virtual ones vanish because $\gamma_{ab} = 0$. The blocks are exact consequences of Eqs. (14), (21) and (26); each of them was generated by substituting the index types into the general formulas and verified numerically (Sec. VII). Every block depends on the vacuum only through $\gamma^{\rm act}$, $\lambda^{\rm act}$ (equivalently $\Gamma^{\rm act}$), f and $\bar{\mathcal{L}}$.

B. Particle–hole channel

Exact identity. The pairs with nonzero norm fall into four sectors, which are the four channels of the MR-RPA of Ref.[5],

$$\text{cv}: (ai), \qquad \text{ca}: (xi), \qquad \text{av}: (ax), \qquad \text{aa}: (xy),\ x \neq y, \tag{30}$$

with their de-excitation partners (ia), (ix), (xa); the pairs (ij) and (ab) have zero norm. The metric is block diagonal in the sectors,

$$\begin{aligned} &\mathcal{S}_{(ai),(bj)} = \delta_{ab}\delta_{ij}, \qquad \mathcal{S}_{(xi),(yj)} = \delta_{ij}(\delta_{xy} - \gamma_{xy}), \\ &\mathcal{S}_{(ax),(by)} = \delta_{ab}\gamma_{xy}, \qquad \mathcal{S}_{(xy),(zw)} = \delta_{xz}\gamma_{yw} - \delta_{yw}\gamma_{xz}, \end{aligned} \tag{31}$$

(in active natural orbitals: 1, $1 - n_x$, n_x and $n_y - n_x$), whereas $\boldsymbol{\mathcal{M}}$ couples all sectors. For cv, ca and av the excitation/de-excitation assignment is fixed, and we list A and B of Eq. (16); for the aa sector it requires the active natural orbitals, so for all blocks involving aa the full $\mathcal{M}_{(pq),(zw)}$ is given for all ordered pairs (zw), from which A and B follow by Eq. (16) after the transformation.

Inactive indices only (cv–cv). The RPA matrices of the core→virtual excitations with the generalized Fock operator; no explicit dependence on the active-space correlation beyond $\gamma^{\rm act}$ in f:

$$A_{(ai),(bj)} = \delta_{ij}f_{ab} - \delta_{ab}f_{ij} + \langle aj||ib\rangle \tag{32}$$

$$B_{(ai),(bj)} = \langle ab||ij\rangle \tag{33}$$

Active indices only (aa–aa). Equation (14) restricted to active indices with $h \to h^{\rm eff}$, *i.e.* the active-space ERPA of `mr-hedin.tex` [its Eq. (erpa-eom)] and of `ekt_erpa.py`:

$$\begin{aligned} M_{(xy),(zw)} = &-\delta_{yw}\bar{\mathcal{L}}_{xz} - \delta_{xz}\bar{\mathcal{L}}_{yw} + f_{xz}\gamma_{yw} \\ &+ f_{yw}\gamma_{xz} - \sum_{uv}\langle uw||vy\rangle\gamma_{ux}\gamma_{vz} - \sum_{uv}\langle uv||yz\rangle\gamma_{ux}\gamma_{vw} \\ &- \sum_{uv}\langle uv||xw\rangle\gamma_{uy}\gamma_{vz} - \sum_{uv}\langle uz||vx\rangle\gamma_{uy}\gamma_{vw} - \tfrac{1}{2}\sum_{uv}\langle uv||yz\rangle\lambda_{uv,xw} \\ &- \tfrac{1}{2}\sum_{uv}\langle uv||xw\rangle\lambda_{uv,yz} + \sum_{uv}\langle uy||vw\rangle\lambda_{ux,vz} + \sum_{uv}\langle ux||vz\rangle\lambda_{uy,vw} \end{aligned} \tag{34}$$

Mixed blocks. Diagonal blocks of the sectors with one active index and all inter-sector couplings:

$$\begin{aligned}
A_{(xi),(yj)} &= \delta_{ij} f_{xy} - \delta_{xy} f_{ij} - \delta_{ij} \bar{\mathcal{L}}_{xy} \\
&\quad + f_{ij}\gamma_{xy} + \langle xj||iy\rangle - \sum_u \langle uj||iy\rangle \gamma_{ux} \\
&\quad + \sum_u \langle ui||xj\rangle \gamma_{uy} - \sum_{uv} \langle uj||vi\rangle \gamma_{ux}\gamma_{vy} + \sum_{uv} \langle ui||vj\rangle \lambda_{ux,vy} \qquad (35)\\
B_{(xi),(yj)} &= \langle xy||ij\rangle - \sum_u \langle uy||ij\rangle \gamma_{ux} + \sum_u \langle ux||ij\rangle \gamma_{uy} \\
&\quad + \sum_{uv} \langle uv||ij\rangle \gamma_{ux}\gamma_{vy} + \tfrac{1}{2} \sum_{uv} \langle uv||ij\rangle \lambda_{uv,xy} \qquad (36)\\
A_{(ax),(by)} &= -\delta_{ab} \bar{\mathcal{L}}_{xy} + f_{ab}\gamma_{xy} - \sum_{uv} \langle ub||va\rangle \gamma_{ux}\gamma_{vy} \\
&\quad + \sum_{uv} \langle ua||vb\rangle \lambda_{ux,vy} \qquad (37)\\
B_{(ax),(by)} &= \sum_{uv} \langle uv||ab\rangle \gamma_{ux}\gamma_{vy} + \tfrac{1}{2} \sum_{uv} \langle uv||ab\rangle \lambda_{uv,xy} \qquad (38)\\
A_{(ai),(xj)} &= \delta_{ij} f_{ax} - \delta_{ij} \bar{\mathcal{L}}_{ax} + \langle aj||ix\rangle \\
&\quad + \sum_u \langle ui||aj\rangle \gamma_{ux} \qquad (39)\\
B_{(ai),(xj)} &= \langle ax||ij\rangle + \sum_u \langle ua||ij\rangle \gamma_{ux} \qquad (40)\\
A_{(ai),(bx)} &= -\delta_{ab} \bar{\mathcal{L}}_{ix} - \sum_u \langle ua||ib\rangle \gamma_{ux} \qquad (41)\\
B_{(ai),(bx)} &= -\sum_u \langle ui||ab\rangle \gamma_{ux} \qquad (42)\\
A_{(xi),(ay)} &= -\sum_u \langle ux||ia\rangle \gamma_{uy} - \sum_{uv} \langle uv||ia\rangle \gamma_{ux}\gamma_{vy} - \tfrac{1}{2} \sum_{uv} \langle uv||ia\rangle \lambda_{uv,xy} \qquad (43)\\
B_{(xi),(ay)} &= \tfrac{1}{2}\delta_{xy} f_{ia} - f_{ia}\gamma_{xy} - \sum_u \langle ui||xa\rangle \gamma_{uy} \\
&\quad + \sum_{uv} \langle ua||vi\rangle \gamma_{ux}\gamma_{vy} - \sum_{uv} \langle ui||va\rangle \lambda_{ux,vy} \qquad (44)\\
M_{(ai),(xy)} &= \sum_u \langle ui||ay\rangle \gamma_{ux} - \sum_u \langle ua||ix\rangle \gamma_{uy} \qquad (45)\\
M_{(xi),(yz)} &= -\delta_{xy} \bar{\mathcal{L}}_{iz} + f_{iz}\gamma_{xy} + \sum_u \langle ui||xz\rangle \gamma_{uy} \\
&\quad - \sum_u \langle ux||iy\rangle \gamma_{uz} - \sum_{uv} \langle uz||vi\rangle \gamma_{ux}\gamma_{vy} - \sum_{uv} \langle uv||iy\rangle \gamma_{ux}\gamma_{vz} \\
&\quad - \tfrac{1}{2} \sum_{uv} \langle uv||iy\rangle \lambda_{uv,xz} + \sum_{uv} \langle ui||vz\rangle \lambda_{ux,vy} \qquad (46)\\
M_{(ax),(yz)} &= -\delta_{xz} \bar{\mathcal{L}}_{ay} + f_{ay}\gamma_{xz} - \sum_{uv} \langle uv||az\rangle \gamma_{ux}\gamma_{vy} \\
&\quad - \sum_{uv} \langle uy||va\rangle \gamma_{ux}\gamma_{vz} - \tfrac{1}{2} \sum_{uv} \langle uv||az\rangle \lambda_{uv,xy} + \sum_{uv} \langle ua||vy\rangle \lambda_{ux,vz} \qquad (47)
\end{aligned}$$

Remarks. (i) In ca–ca and av–av the bare integral is dressed by the active hole density $\delta_{xy} - \gamma_{xy}$ and by γ, respectively, and the cumulant enters through $\langle ui||vj\rangle\lambda_{ux,vy}$, $\frac{1}{2}\langle uv||ij\rangle\lambda_{uv,xy}$ and $\bar{\mathcal{L}}_{xy}$ (analogously with a, b). The combinations $\gamma_{ux}\gamma_{vy} + \frac{1}{2}\lambda_{uv,xy}$ contracted with an integral antisymmetric in u, v equal $\frac{1}{2}\Gamma_{uv,xy}$, so the B-type and ca–av blocks can equally be written with the active 2-RDM. (ii) The couplings between the inactive sector cv and the active-containing sectors contain no cumulant: cv–ca and cv–av involve only γ^{act} and $\bar{\mathcal{L}}$, cv–aa only γ^{act}. The cumulant first appears in couplings with at least two active indices in the ket pair or two active indices across the two pairs. (iii) For an empty active space only Eqs. (32)–(33) survive. For a determinantal $|\Xi\rangle$ ($\lambda = 0$, γ^{act} idempotent) every block reduces to the corresponding Hartree–Fock RPA block or vanishes together with its metric. (iv) The MR-RPA of Ref.[5] screens with the direct residual kernel v^R contracted with the exact channel transition densities

and keeps the exact active response in aa; the blocks above are instead the single-excitation-manifold double commutator with the antisymmetrized bare kernel in all four sectors, *i.e.* the extended form of the fixed-kernel model of `Pa2CCD.tex`. The two coincide in cv–cv (up to direct versus antisymmetrized kernel) and differ elsewhere.

C. Particle–particle channel

Exact identity. The pairs with nonzero norm fall into five sectors,

$$\begin{aligned} &\text{vv}: (ab), \qquad \text{av}: (ax), \qquad \text{aa}: (xy),\ x<y, \\ &\qquad\qquad\qquad\quad \text{ca}: (ix), \qquad \text{cc}: (ij), \end{aligned} \tag{48}$$

while the core–virtual pairs (ai) have zero norm and drop out. The metric is block diagonal,

$$\begin{aligned} \mathcal{S}^{\text{pp}}_{(ab),(cd)} &= \delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}, \qquad \mathcal{S}^{\text{pp}}_{(ax),(by)} = \delta_{ab}(\delta_{xy} - \gamma_{xy}), \\ \mathcal{S}^{\text{pp}}_{(ix),(jy)} &= -\delta_{ij}\gamma_{xy}, \qquad \mathcal{S}^{\text{pp}}_{(ij),(kl)} = -(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}), \\ \mathcal{S}^{\text{pp}}_{(xy),(zw)} &= \delta_{xz}\delta_{yw} - \delta_{xw}\delta_{yz} - \delta_{xz}\gamma_{yw} + \delta_{xw}\gamma_{yz} \\ &\quad + \delta_{yz}\gamma_{xw} - \delta_{yw}\gamma_{xz}, \end{aligned} \tag{49}$$

(in active natural orbitals: $1,\ 1-n_x,\ 1-n_x-n_y,\ -n_x,\ -1$), so that vv and av belong to the particle-like sector $\mathcal{P}$, cc and ca to the hole-like sector $\mathcal{H}$, and aa splits according to the sign of $1-n_x-n_y$. The fifteen blocks of $\boldsymbol{\mathcal{M}}^{\text{pp}}$ are ($P(ab)$ etc. are written out):

Inactive indices only (vv–vv, cc–cc, vv–cc). The Hartree–Fock particle–particle, hole–hole and coupling matrices with the generalized Fock operator [$B^{\text{pp}}_{ab,ij} = -\mathcal{M}^{\text{pp}}_{(ab),(ij)} = \langle ab||ij\rangle$]:

$$\begin{aligned} M^{\text{pp}}_{(ab),(cd)} &= \delta_{bd}f_{ac} - \delta_{bc}f_{ad} - \delta_{ad}f_{bc} \\ &\quad + \delta_{ac}f_{bd} + \langle ab||cd\rangle \end{aligned} \tag{50}$$

$$\begin{aligned} M^{\text{pp}}_{(ij),(kl)} &= -\delta_{jl}f_{ik} + \delta_{jk}f_{il} + \delta_{il}f_{jk} \\ &\quad - \delta_{ik}f_{jl} + \langle ij||kl\rangle \end{aligned} \tag{51}$$

$$M^{\text{pp}}_{(ab),(ij)} = -\langle ab||ij\rangle \tag{52}$$

Active indices only (aa–aa). Equation (21) restricted to active indices with $h \to h^{\text{eff}}$, *i.e.* the particle–particle ERPA of the active-space problem:

$$\begin{aligned} M^{\text{pp}}_{(xy),(zw)} &= \delta_{yw}f_{xz} - \delta_{yz}f_{xw} - \delta_{xw}f_{yz} \\ &\quad + \delta_{xz}f_{yw} - \delta_{yw}\bar{\mathcal{L}}_{xz} + \delta_{yz}\bar{\mathcal{L}}_{xw} \\ &\quad + \delta_{xw}\bar{\mathcal{L}}_{yz} - \delta_{xz}\bar{\mathcal{L}}_{yw} - f_{xz}\gamma_{yw} \\ &\quad + f_{xw}\gamma_{yz} + f_{yz}\gamma_{xw} - f_{yw}\gamma_{xz} \\ &\quad + \langle xy||zw\rangle - \sum_u \langle uy||zw\rangle\gamma_{ux} + \sum_u \langle ux||zw\rangle\gamma_{uy} \\ &\quad - \sum_u \langle uw||xy\rangle\gamma_{uz} + \sum_u \langle uz||xy\rangle\gamma_{uw} + \sum_{uv} \langle uy||vw\rangle\gamma_{ux}\gamma_{vz} \\ &\quad - \sum_{uv} \langle uy||vz\rangle\gamma_{ux}\gamma_{vw} - \sum_{uv} \langle ux||vw\rangle\gamma_{uy}\gamma_{vz} + \sum_{uv} \langle ux||vz\rangle\gamma_{uy}\gamma_{vw} \\ &\quad - \sum_{uv} \langle uw||vy\rangle\lambda_{ux,vz} + \sum_{uv} \langle uz||vy\rangle\lambda_{ux,vw} + \sum_{uv} \langle uw||vx\rangle\lambda_{uy,vz} \\ &\quad - \sum_{uv} \langle uz||vx\rangle\lambda_{uy,vw} \end{aligned} \tag{53}$$

Mixed blocks.

$$
\begin{aligned}
M^{\mathrm{pp}}_{(ax),(by)} &= \delta_{xy} f_{ab} + \delta_{ab} f_{xy} - \delta_{ab} \bar{\mathcal{L}}_{xy} \\
&\quad - f_{ab}\gamma_{xy} + \langle ax \| by \rangle + \sum_u \langle ua \| by \rangle \gamma_{ux} \\
&\quad + \sum_u \langle ub \| ax \rangle \gamma_{uy} + \sum_{uv} \langle ua \| vb \rangle \gamma_{ux}\gamma_{vy} - \sum_{uv} \langle ub \| va \rangle \lambda_{ux,vy} \qquad (54) \\
M^{\mathrm{pp}}_{(ix),(jy)} &= -\delta_{ij} \bar{\mathcal{L}}_{xy} - f_{ij}\gamma_{xy} + \sum_{uv} \langle ui \| vj \rangle \gamma_{ux}\gamma_{vy} \\
&\quad - \sum_{uv} \langle uj \| vi \rangle \lambda_{ux,vy} \qquad (55) \\
M^{\mathrm{pp}}_{(ab),(cx)} &= -\delta_{bc} f_{ax} + \delta_{ac} f_{bx} + \delta_{bc} \bar{\mathcal{L}}_{ax} \\
&\quad - \delta_{ac} \bar{\mathcal{L}}_{bx} + \langle ab \| cx \rangle + \sum_u \langle uc \| ab \rangle \gamma_{ux} \qquad (56) \\
M^{\mathrm{pp}}_{(ab),(xy)} &= \langle ab \| xy \rangle - \sum_u \langle uy \| ab \rangle \gamma_{ux} + \sum_u \langle ux \| ab \rangle \gamma_{uy} \qquad (57) \\
M^{\mathrm{pp}}_{(ab),(ix)} &= \sum_u \langle ui \| ab \rangle \gamma_{ux} \qquad (58) \\
M^{\mathrm{pp}}_{(ax),(yz)} &= \delta_{xz} f_{ay} - \delta_{xy} f_{az} - \delta_{xz} \bar{\mathcal{L}}_{ay} \\
&\quad + \delta_{xy} \bar{\mathcal{L}}_{az} - f_{ay}\gamma_{xz} + f_{az}\gamma_{xy} \\
&\quad + \langle ax \| yz \rangle + \sum_u \langle ua \| yz \rangle \gamma_{ux} - \sum_u \langle uz \| ax \rangle \gamma_{uy} \\
&\quad + \sum_u \langle uy \| ax \rangle \gamma_{uz} - \sum_{uv} \langle ua \| vz \rangle \gamma_{ux}\gamma_{vy} + \sum_{uv} \langle ua \| vy \rangle \gamma_{ux}\gamma_{vz} \\
&\quad + \sum_{uv} \langle uz \| va \rangle \lambda_{ux,vy} - \sum_{uv} \langle uy \| va \rangle \lambda_{ux,vz} \qquad (59) \\
M^{\mathrm{pp}}_{(ax),(iy)} &= \tfrac{1}{2}\delta_{xy} f_{ai} - f_{ai}\gamma_{xy} + \sum_u \langle ui \| ax \rangle \gamma_{uy} \\
&\quad + \sum_{uv} \langle ua \| vi \rangle \gamma_{ux}\gamma_{vy} - \sum_{uv} \langle ui \| va \rangle \lambda_{ux,vy} \qquad (60) \\
M^{\mathrm{pp}}_{(ax),(ij)} &= -\langle ax \| ij \rangle - \sum_u \langle ua \| ij \rangle \gamma_{ux} \qquad (61) \\
M^{\mathrm{pp}}_{(xy),(iz)} &= \delta_{yz} f_{xi} - \delta_{xz} f_{yi} - \delta_{yz} \bar{\mathcal{L}}_{xi} \\
&\quad + \delta_{xz} \bar{\mathcal{L}}_{yi} - f_{xi}\gamma_{yz} + f_{yi}\gamma_{xz} \\
&\quad + \sum_u \langle ui \| xy \rangle \gamma_{uz} - \sum_{uv} \langle uy \| vi \rangle \gamma_{ux}\gamma_{vz} + \sum_{uv} \langle ux \| vi \rangle \gamma_{uy}\gamma_{vz} \\
&\quad + \sum_{uv} \langle ui \| vy \rangle \lambda_{ux,vz} - \sum_{uv} \langle ui \| vx \rangle \lambda_{uy,vz} \qquad (62) \\
M^{\mathrm{pp}}_{(xy),(ij)} &= -\langle xy \| ij \rangle + \sum_u \langle uy \| ij \rangle \gamma_{ux} - \sum_u \langle ux \| ij \rangle \gamma_{uy} \qquad (63) \\
M^{\mathrm{pp}}_{(ix),(jk)} &= \delta_{ik} \bar{\mathcal{L}}_{xj} - \delta_{ij} \bar{\mathcal{L}}_{xk} - \sum_u \langle ui \| jk \rangle \gamma_{ux} \qquad (64)
\end{aligned}
$$

Remarks. (i) The ladder integral is dressed on each leg by $-\gamma$ [Eq. (20)]; a core leg therefore subtracts one copy of the bare integral, so that two core legs in a pair give $\mathcal{M}^{\mathrm{pp}}_{(ab),(ij)} = -\langle ab \| ij \rangle$ and one core leg gives the $-\langle \cdot\cdot \| i \cdot \rangle$ combinations, while virtual legs are undressed. (ii) The cumulant enters only through the particle–hole-type contractions $\langle u \cdot \| v \cdot \rangle \lambda_{u\cdot,v\cdot}$ and through $\bar{\mathcal{L}}$; it is absent from all blocks whose ket or bra pair is purely inactive. (iii) The cv pairs are absent from the manifold but the av and ca sectors couple the particle-like and hole-like spaces through the active density: $\mathcal{M}^{\mathrm{pp}}_{(ax),(iy)}$ is the extended-vacuum analogue of a $\boldsymbol{B}^{\mathrm{pp}}$ block with one active index on each side.

D. Crossed-ring channel

Proposed approximation. By Eq. (25) the crossed-ring channel has the same sectors, the same metric blocks (31) and the same A blocks as the particle–hole channel, while every B block changes sign, $B^{\overline{\mathrm{eh}}} = -B$, including the B blocks of

the aa sector obtained from $\mathcal{M}_{(pq),(zw)}$ in the active natural-orbital basis. Equivalently, $\boldsymbol{\mathcal{M}}^{\overline{\text{eh}}} = \boldsymbol{Z}\boldsymbol{\mathcal{M}}\boldsymbol{Z}$ with $\boldsymbol{Z} = +1$ on $(ai), (xi), (ax)$ and on the active pairs with $n_y > n_x$, and $\boldsymbol{Z} = -1$ on their partners; $\boldsymbol{Z}$ acts within each sector, so the sector structure is preserved.

VII. NUMERICAL VERIFICATION

All equations of Secs. III–VI were checked against brute-force double commutators evaluated with explicit second-quantized operators in the full Fock space (`numerics/erpa_blocks/`):

(a) Eqs. (12)–(15) and (19)–(21), including the unsymmetrized forms, Eq. (7), the identity $\mathcal{M}_{(qp),(sr)} = \mathcal{M}_{(pq),(rs)}$ and the cumulant form of $\mathcal{L}$, for random real h, $(pr|qs)$ with the eightfold symmetry and a random (non-eigenstate) three-electron vector in seven spin orbitals: maximum deviations $\leq 2 \times 10^{-15}$ (`verify_general.py`).

(b) Eqs. (26)–(28), (31), (49) and all 35 blocks of Secs. VI B–VI C, for a product state with two core, four active (two electrons, random CI vector) and two virtual spin orbitals and random integrals, so that $\mathcal{L}$ is not symmetric: maximum deviations $\leq 3 \times 10^{-15}$ (`blocks.py`, which also generates the LaTeX of the blocks from Eqs. (14) and (21) by substituting the index types).

The Hartree–Fock limits quoted in Secs. III–V are the cv–cv, vv–vv, cc–cc and vv–cc blocks with $\gamma^{\text{act}} = 0$.

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